MIL-P-60017A (PA)
22 May 75
SUPERSEDING
MIL-P-60017 (MU)
30 January 1964

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M60A1 PARTS FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts for one type of percussion primer designated as Primer, Percussion, M60A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1168	- Ammunition Lot Numbering
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

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DRAWINGS

ARMY

7548485 - Primers, Percussion and Igniting. Details, Liners
7548516 - Primers, Percussion and Igniting. Details, Diaphragms
8838086 - Head
8838089 - Body Part No. 8838089-19

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Part.-The parts shall comply with all requirements specified on Drawings (dwgs.) 7548485, 7548516, 8838086 and 8838089 as specified in applicable specifications.

3.3 Hydrostatic Test.-The primer tubing shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws, as evidenced by leaks, and without permanent deformation beyond drawing requirements, when tested as specified in 4.4.1 (see 6.3).

3.4 Workmanship.-All parts shall be dry and free of cracks, chips, dust, grease, rust, burrs and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.5 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 25 each individual components, which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirement of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes, and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample or portion thereof as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspection to be performed.-Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of assembly component in the sample to comply with any or all of the stated requirements.

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4.3 Inspection Provisions

4.3.1 Inspection Lot Formation.-The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Lot numbering, as required. shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Sampling plans shall be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Liner (see dwg. 7548485).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:	None defined	
Minor:		
	201. Length	Gage
	202. Width	Gage

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4.3.2.2 Diaphragm (see dwg. 7548516).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:	None defined	
Minor:		
	201. Large outside diameter	Gage
	202. Small outside diameter	Gage
	203. Total length	Gage

4.3.2.3 Head (see dwg. 8838086).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
	101. Major diameter of thread, min.	Gage
	102. Pitch diameter of thread, min.	Gage
	103. Concentricity of primer element cavity with body diameter	Gage
	104. Concentricity of flange diameter with body diameter	Gage
	105. Concentricity of flash hole with primer element cavity	Gage
	106. Diameter of primer element cavity, max.	Gage
	107. Diameter of flash hole	Gage
	108. Depth to bottom of primer element cavity	Gage
	109. Diameter of body	Gage
	110. Flash hole missing or obstructed	Visual
Minor:		
	201. Major diameter of thread, max.	Gage
	202. Pitch diameter of thread, max.	Gage
	203. Outside diameter of forward end.	Gage

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204. Depth of forward end cavity.....Gage
205. Width of undercut, max. .....Gage
206. Length of thread including undercut, min. .....Gage
207. Distance from undercut to forward end .....Gage
208. Total length, min. .....Gage
209. Thickness of flange, min. .....Gage
210. Diameter of flange, min. .....Gage
211. Diameter of forward cavity .....Gage
212. Finish improper .....Visual
213. Evidence of poor workmanship (see 3.4) .....Visual

4.3.2.4 Body (see dwg. 8838089).

Categories	Defects	Method of Inspection
Critical:		
	1. Flash hole missing	Gage
	2. Crack or split	Visual
Major:		
	101. Pitch diameter of thread, max.	Gage
	102. Minor diameter of thread, max.	Gage
	103. Outside diameter, max.	Gage
	104. Wall thickness, min.	Gage
	105. Diameter of holes	Gage
	106. Location of holes	Gage
	107. Depth to which plug gage must enter	Gage
Minor:		
	201. Pitch diameter of thread	Gage
	202. Depth of counterbore, max.	Gage
	203. Wall thickness, max.	Gage
	204. Diameter of counterbore	Gage
	205. Length of thread, min.	Gage
	206. Protective finish with bare spot	Visual
	207. Evidence of poor workmanship (see 3.4)	Visual

4.3.3 Testing

4.3.3.1 Hydrostatic test (see 3.3) Critical defect - This test shall be conducted 100 percent. If any primer

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tubing fails to comply with the applicable requirement, it shall be classed defective and no primer bodies shall be fabricated therefrom. The test shall be performed as specified in 4.4.1 using equipment specified in 4.3.4.

4.3.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and 4.3.3 (see 6.2).

4.4 Test Methods and Procedures

4.4.1 Hydrostatic Test.-Prior to machining, the primer tubing shall be tested for not less than five seconds at the minimum hydrostatic pressure as calculated by the following formula:

$$P = \frac{2\ St}{D} X.9$$

where:

P = test pressure in pounds per square inch (psi.).
S= minimum specified yield strength, psi.
t = specified maximum wall thickness in inches.
D = specified maximum outside diameter in inches.

Observation shall be made for rupture, or indication of flaws as evidenced by leaks, and for permanent deformation beyond drawing requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-The parts shall be packaged, packed and marked in accordance with Standard MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. First Article Inspection provisions.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are

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designated by drawing numbers under the "Method of Inspection" heading Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204 "Inspection Equipment Design" for guidance. The item detail specification number, paragraph number, and defect number from Section 4 shall be referenced on each contractor designs together with the component or assembly drawing number, revision letter and date to which the specific design applies.

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6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Hydrostatic Test.-If the primer tubing procured for the bodies manufactured in accordance with this specification has been previously subjected to the hydrostatic test specified in this specification a second hydrostatic test will not be required.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1390-A103

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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